

AI FITNESS & WELLNESS PLATFORM

Full Platform Architecture Plan — Phase 0 through Phase 4

Foundation → Business MVP → Coaching Layer → AI Layer → Advanced AI

1. Executive Summary

This document defines the full architecture plan for all five build phases of the AI Fitness & Wellness Platform — Phase 0 (Foundation), Phase 1 (Business Platform MVP), Phase 2 (Coaching Layer), Phase 3 (AI Layer), and Phase 4 (Advanced AI). It is derived from the full 60-section product scope, structured into a buildable, sequenced plan suitable for stakeholder sign-off.

The platform is designed around one core idea: a closed-loop system that continuously Assesses a member, Plans for them, watches them Train, Observes and measures the outcome, and Adapts the next plan. Every phase below — including the AI phases — is explained with what gets built, why it's sequenced where it is, and what "done" looks like.

Key takeaway for this review

- This plan covers the entire platform vision end-to-end — not just the early phases. Each phase has its own architecture diagram, module breakdown, rationale, and exit criteria.
- Phases are still built in order: Phase 3-4 (AI, computer vision, voice AI) technically depend on data and workflows established in Phase 0-2, so they are sequenced later even though they are fully architected here.
- The architecture is designed so each phase builds cleanly on the last, without re-architecting earlier work — the AI Gateway (Phase 3) and event history (Phase 0) are the two design choices that make this possible.

2. Guiding Principles

Four architectural decisions shape everything in this plan. These are the points most worth discussing in the review, since they affect cost, timeline, and flexibility.

2.1 The Closed-Loop Model

Assess → Plan → Train → Observe → Adapt. Phase 0-2 builds the data and workflow foundation for this loop; Phase 3-4 adds the AI that closes it automatically.

2.2 Member 360 as the Single Source of Truth

Every module — CRM, billing, assessments, programs, nutrition — writes into one unified member data profile. This is a database design principle, not a UI screen: it is what allows AI features later to reason over one consistent picture of a member instead of stitching together disconnected tables.

2.3 Modular Monolith, Not Microservices (initially)

The platform will be built as a single deployable application, internally organized into clean, independent modules (billing, scheduling, assessments, etc.). This avoids the operational overhead of microservices (separate deployments, service-to-service networking, distributed monitoring) that isn't justified at this stage. Modules are still designed with clear boundaries so specific pieces (e.g. computer vision inference in Phase 4) can be split into their own service later without a rewrite.

2.4 Multi-Tenancy & Event Logging from Day One

Every table in the system is designed from the start to isolate data per client organization (tenant), and every meaningful action (missed workout, payment, assessment completed) is logged as an event. Both are extremely expensive to retrofit later, and event history logged early becomes training/reference data for the AI features in Phase 3-4.

3. System Architecture Overview

The platform is organized into three towers — Business Platform, Coaching Platform, and AI Platform — all reading from and writing to one Member 360 profile, which sits on top of a shared set of core services (multi-tenancy, authentication/RBAC, event bus, and the primary database).

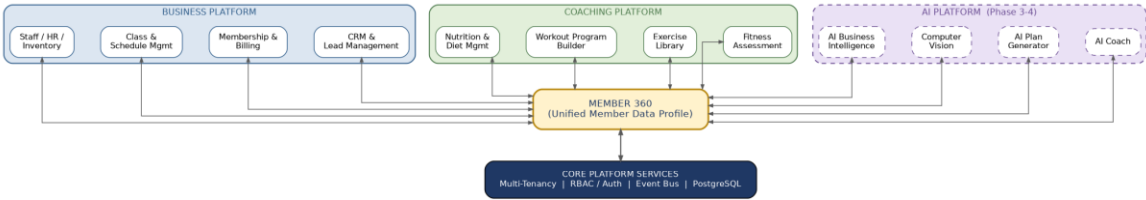


Figure 1 — High-level system architecture. The AI Platform (dashed) is delivered in Phase 3-4, after Business and Coaching Platform are live.

This diagram is the one worth spending the most time on in the review meeting — it shows the entire platform vision across all five phases, with every tower converging on Member 360 and the same Core Platform Services underneath. The dashed styling on the AI Platform simply reflects build order, not planning depth — Phase 3-4 are detailed in full later in this document.

4. Technology Stack

Layer	Technology	Why
Backend framework	Django + Django REST Framework	Mature, batteries-included, fast to build secure CRUD + APIs; strong ORM for complex relational data
Database	PostgreSQL	Relational integrity for tenant/member data; strong support for JSON fields (useful for flexible assessment data) and row-level isolation patterns
API style	REST, API-first	Every feature exposed via API from day one so web, mobile, and future AI services can all consume the same backend
Frontend (web)	React	Component reusability, large ecosystem, matches existing team experience
Mobile	Flutter	Single codebase for iOS + Android member-facing app
Background jobs / events	Celery + Redis (or equivalent)	Async processing for notifications, reminders, and the event bus
Auth	JWT-based auth + RBAC middleware	Stateless auth suited to API-first + mobile; RBAC enforced at the API layer

Layer	Technology	Why
Hosting	Cloud (AWS/GCP/Azure — TBD)	Scalable, supports later addition of GPU-backed services for CV/AI in Phase 3-4
AI / LLM (Phase 3)	AI Gateway + external LLM API(s)	Powers AI Coach and plan generation; gateway keeps the platform vendor-agnostic
Retrieval layer (Phase 3)	Vector DB (e.g. pgvector on Postgres)	Lets the AI Coach retrieve relevant Member 360 context before answering (RAG pattern)
Computer Vision (Phase 4)	Pose-estimation model (e.g. MediaPipe/OpenPose), edge-first inference	Real-time form feedback; processing kept close to the device for speed and privacy
Predictive models (Phase 4)	Lightweight ML (e.g. gradient boosting) on event history	Churn/retention scoring — doesn't need deep learning to be useful at this stage
Voice AI (Phase 4)	Telephony API + speech-to-text/text-to-speech + LLM	Powers the AI Sales Agent for calls; a distinct stack from the text-based AI Coach

5. Phase Roadmap

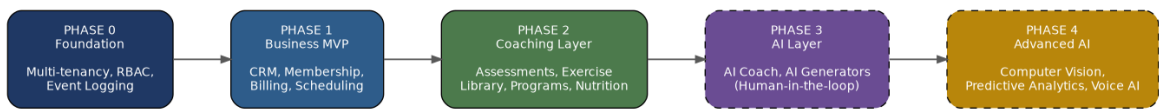


Figure 2 — Five-phase roadmap. Every phase below is detailed in full: diagram, modules, rationale, and exit criteria.

Phase	Goal	Depends on AI?
Phase 0 — Foundation	Multi-tenancy, RBAC, core data model, event logging	No
Phase 1 — Business MVP	CRM, membership, billing, scheduling — a sellable product	No
Phase 2 — Coaching Layer	Assessments, exercise library, manual program & diet builder	No
Phase 3 — AI Layer	AI Coach, AI-generated plans (trainer-approved), AI Gateway live	Yes — requires Phase 0-2 data
Phase 4 — Advanced AI	Computer vision form-check, churn prediction, voice AI sales agent	Yes — requires usage history from Phase 1-3

6. Phase 0 — Foundation

Purpose: establish the structural decisions that are extremely costly to change later — before any customer-facing feature is built.

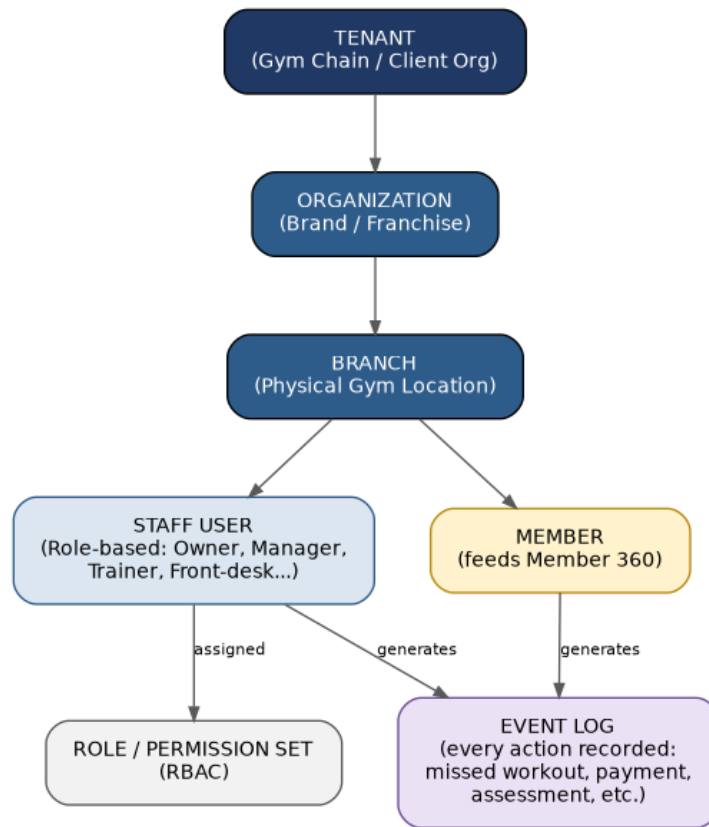


Figure 3 — Tenancy, RBAC and event-logging hierarchy established in Phase 0.

6.1 What gets built

- Tenant → Organization → Branch data hierarchy, with every downstream table scoped to a tenant
- Role-Based Access Control (RBAC): role and permission framework covering staff roles (Owner, Manager, Trainer, Front-desk, etc.)
- Core Member entity — the seed of the future Member 360 profile
- Event logging system: a central event bus/table that records meaningful actions across the platform
- Authentication (JWT-based), password policies, session handling
- Base API scaffolding and environment setup (dev/staging/prod)

6.2 Why this order matters

These four items — tenancy, RBAC, Member 360 seed, and event logging — are foundational because nearly every table and feature built afterward references them. Retrofitting tenant isolation or event logging into a system that already has hundreds of tables and business rules is one of the most expensive refactors in software; doing it first avoids that cost entirely.

6.3 Exit criteria for Phase 0

- A new tenant/organization/branch can be provisioned end-to-end
- Staff users can be created and assigned roles with enforced permissions
- A test event can be logged and queried against a member record
- Core API layer is live with authentication working

7. Phase 1 — Business Platform MVP

Purpose: deliver a complete, independently sellable gym/business management product — no AI required. This is the commercial foundation.

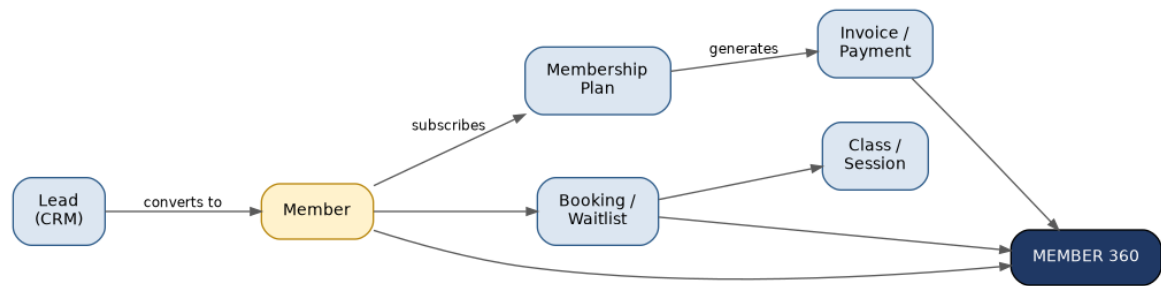


Figure 4 — Core Phase 1 module relationships, all converging on Member 360.

7.1 What gets built

Module	Core capability
CRM & Lead Management	Capture leads, track follow-ups, convert leads to paying members
Membership Management	Plans, pricing, renewals, freezes, upgrades/downgrades
Billing & Payments	Invoicing, payment gateway integration, trainer commission tracking
Class & Schedule Management	Class creation, booking, waitlists, trainer assignment
Staff / HR / Inventory (light)	Trainer attendance, basic stock tracking for the gym

7.2 Why this comes before the AI/coaching features

A gym cannot operate without membership, billing, and scheduling — these are needed regardless of whether any AI feature ever ships. Building this first also means the platform has real, sellable value early, independent of how the AI phases progress.

7.3 Exit criteria for Phase 1

- A lead can be captured, followed up, and converted into a paying member
- A member can be billed, and a class can be booked and attended
- All actions from this phase are visible in the Member 360 timeline

8. Phase 2 — Coaching Layer

Purpose: add the fitness/coaching content and workflows — still without AI. A human trainer builds and assigns everything; the system just structures and stores it well.

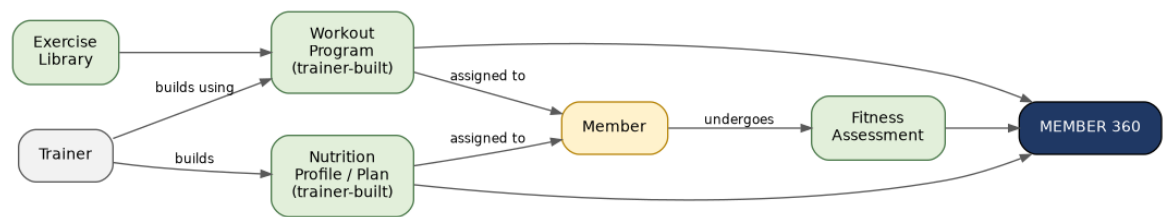


Figure 5 — Coaching layer module relationships. All AI in this diagram is deferred to Phase 3.

8.1 What gets built

Module	Core capability
Fitness Assessment	Structured intake: measurements, body composition, mobility, strength baselines
Exercise Library	Searchable database of exercises with instructions, media, difficulty, muscle groups
Workout Program Builder	Trainer assembles exercises into weekly/monthly programs, assigned to a member
Nutrition & Diet Management	Trainer-built meal plans and macro/calorie targets, including an Indian food dataset

8.2 Why this comes before AI generation

The AI Plan Generator (Phase 3) needs a well-structured Exercise Library and assessment data to generate anything useful — an AI cannot generate a good workout plan without a real data set to build from. Phase 2 is what makes Phase 3 possible: it's the content and workflow layer the AI will eventually sit on top of and accelerate.

8.3 Exit criteria for Phase 2

- A trainer can run a full assessment on a member and see it saved to Member 360
- A trainer can build a program from the Exercise Library and assign it to a member
- A trainer can build a diet plan and assign it to a member

9. Phase 3 — AI Layer

Purpose: switch on the first AI-powered features — an AI Coach and AI-generated workout/diet plans — built on top of the data and content created in Phase 0-2. This is where the AI Gateway becomes a real, running component rather than just an architectural placeholder.

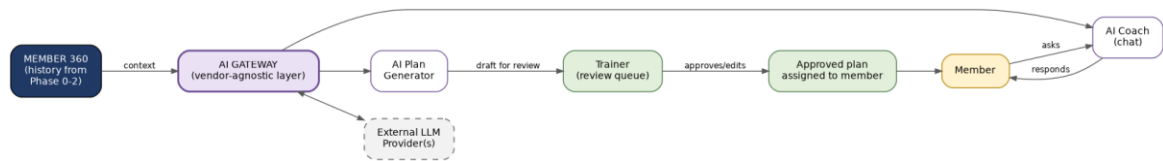


Figure 7 — Phase 3 flow: Member 360 context feeds the AI Gateway, which powers the AI Coach and Plan Generator, with a trainer approval step before anything reaches the member.

9.1 What gets built

Module	Core capability
AI Gateway	Central abstraction layer that routes requests to one or more LLM providers, so the rest of the platform never talks to a vendor API directly
AI Coach	Conversational assistant members can ask fitness/nutrition questions to, using their own Member 360 data as context
AI Plan Generator	Drafts a workout or diet plan using the Phase 2 Exercise Library and a member's assessment data
Trainer Review Queue	Every AI-generated plan lands here for a trainer to approve, edit, or reject before it reaches a member

9.2 Why this comes after Phase 0-2, not before

The AI Coach and Plan Generator are only as good as the data they draw on. Without a populated Exercise Library, structured assessments, and real Member 360 history from Phase 0-2, the AI has nothing meaningful to reason over — it would either produce generic, low-value output or need to be rebuilt once real data exists. Building the AI Gateway now, on top of a stable foundation, also means it's built once, correctly, rather than retrofitted.

9.3 Human-in-the-loop is a hard requirement, not a preference

No AI-generated plan reaches a member without a trainer approval step. This directly reflects the scope document's own three-tier model: AI can auto-execute low-risk actions (reminders, content suggestions), AI can recommend but a human must approve (workout/diet changes), and some decisions remain human-only (anything medical). Phase 3 architecture enforces this at the workflow level — the review queue is not optional and cannot be bypassed by configuration.

9.4 Exit criteria for Phase 3

- The AI Gateway can route a request to at least one LLM provider and return a response
- A member can ask the AI Coach a question and get a context-aware answer grounded in their own Member 360 data
- An AI-generated plan can be created, reviewed, edited, and approved by a trainer before assignment

- Switching the underlying LLM provider requires a configuration change, not a code change in the Coach or Generator modules

10. Phase 4 — Advanced AI

Purpose: add the most differentiated — and most technically demanding — capabilities: computer vision form-checking, predictive retention modeling, and a voice AI sales agent. These are sequenced last because each one depends on data, infrastructure, or trust built in earlier phases.

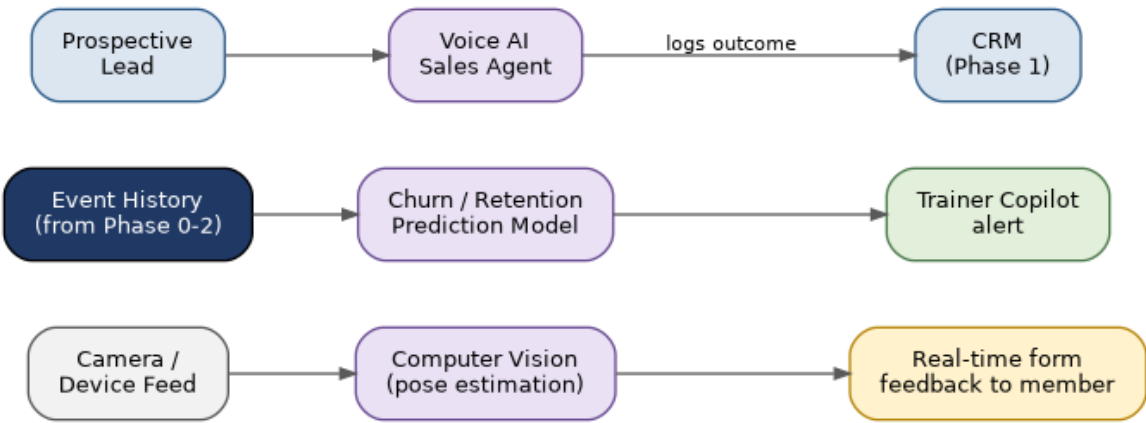


Figure 8 — Phase 4's three largely independent capabilities: computer vision, predictive analytics, and voice AI — each consuming data or infrastructure from earlier phases.

10.1 What gets built

Module	Core capability
Computer Vision (form analysis)	Uses a camera feed and pose-estimation to give a member real-time feedback on exercise form (e.g. “knees caving in”)
Predictive Analytics (churn/retention)	Scores members on likelihood to disengage, using event history accumulated since Phase 0, and alerts trainers via Copilot
Voice AI Sales Agent	Places or receives calls with prospective leads, qualifies them conversationally, and logs the outcome back into the Phase 1 CRM

10.2 Why this comes last

Each Phase 4 capability has a hard dependency that only exists after earlier phases are live:

- Computer vision needs the Exercise Library (Phase 2) to know what “correct form” looks like for a given movement
- Churn prediction needs months of real event history (Phase 0 onward) — a model trained on no data isn't useful, and this history cannot be created retroactively
- The voice AI sales agent needs the CRM (Phase 1) to log into, and benefits from the AI Gateway (Phase 3) already being a proven, stable pattern in production

10.3 Special considerations

- Computer vision inference is planned as edge-first (processed close to the member's device) rather than sent to the cloud, for both latency and privacy reasons
- Camera and biometric data require explicit, purpose-specific consent — this is a DPDP requirement, not just good practice (see Section 12.2)

- Voice AI calling should be scoped against telecom/consent regulations for outbound calls in each operating region before build

10.4 Exit criteria for Phase 4

- A member's exercise form can be analyzed live and specific, actionable feedback surfaced during the movement
- A churn-risk score is generated per member and surfaced to a trainer with enough lead time to act
- A voice AI call with a lead can be completed end-to-end, with the outcome correctly logged in the CRM

11. Why Event Logging Starts Now: A Worked Example

This is a concrete illustration of why Phase 0's event logging matters even though no AI exists yet in Phase 0-2.



Figure 6 — An event logged in Phase 0-2 becomes usable AI input in Phase 3-4, but only if it was captured at the time.

If “workout_missed” events aren't logged from day one, there is no historical record for a future churn-prediction model to learn from — that history cannot be recreated retroactively. Every month of Phase 1-2 operation without event logging is a month of lost training data for Phase 3-4.

12. Cross-Cutting Concerns

12.1 Security (built into every phase)

- TLS in transit, encryption at rest for sensitive fields
- RBAC enforced at the API layer, not just the UI
- Tenant isolation enforced at the query layer (not just application logic)
- Full audit logging of access to sensitive member data

12.2 DPDP Act, 2023 — Status

The architecture supports the core DPDP concepts — consent capture, data minimization, retention rules, and data export/deletion — as part of the Member 360 and event design. However, the following statutory items are not yet addressed anywhere in the current scope and should be flagged as follow-up items, ideally with legal input, before public launch — several of these become urgent specifically once Phase 4 (camera/biometric data, voice calling) is reached:

- Breach notification process (to the Data Protection Board and affected users)
- Cross-border data transfer rules (relevant once any AI vendor, e.g. a US-based LLM API, or cloud host is chosen — relevant from Phase 3 onward)
- Grievance Officer / DPO designation for data-rights complaints specifically (distinct from general customer support)
- Parental consent handling for members under 18
- Explicit, purpose-specific consent for camera/biometric data ahead of Phase 4 computer vision

12.3 SOC 2 — Status

SOC 2 is not a certification a document or architecture plan can grant — it requires an independent auditor to verify security controls operate correctly over an extended period (often 3-12 months of live evidence). This architecture is designed to make a future SOC 2 audit achievable (RBAC, encryption, tenant isolation, audit logs are all in scope from Phase 0), but formal SOC 2 pursuit should be treated as a later milestone, typically triggered by enterprise/B2B customer requirements rather than a phase-specific deliverable.

12.4 AI Gateway — Design Now, Activate in Phase 3

The AI Gateway is fully designed as part of this plan (Section 9.1) but only becomes an active, running component in Phase 3. The module boundaries in Phase 1-2 are deliberately kept clean specifically so the Gateway can be introduced without reworking earlier modules — this is why it's architected alongside Phase 0-2 even though it isn't switched on until later.

13. Phase Sequencing Rationale

Every capability in the original 60-section scope is represented somewhere in Phase 0-4 above — nothing has been dropped. What this plan does is impose a build order based on real dependencies, so the team isn't asked to build AI features before there's data for them to use, or computer vision before there's an Exercise Library to validate form against. The table below summarizes why each AI/advanced capability sits where it does.

Capability	Phase	Hard dependency
AI Coach / chatbot	3	Member 360 history (Phase 0-2) to give it something to reason over
AI-generated workout/diet plans	3	Exercise Library + assessments (Phase 2) as source content
Computer vision form analysis	4	Exercise Library (Phase 2) defines what “correct form” means
Churn / retention prediction	4	Months of real event history (Phase 0 onward) to train against
AI Sales Agent / voice AI	4	CRM (Phase 1) to log into; proven AI Gateway pattern (Phase 3)
AI Business Intelligence (NL queries)	4	Sufficient operational data across all prior phases to query meaningfully

14. Risks & Dependencies

Risk	Mitigation
Multi-tenancy design flaws surface late	Dedicated schema review before any Phase 1 module starts; automated tests for tenant data isolation
Scope creep — AI/CV work starting during Phase 1-2	Explicit phase gate reviews; Phase 3-4 work not started until Phase 0-2 exit criteria are met
DPDP gaps discovered post-launch	Legal review of data flows scheduled before public launch, and specifically before Phase 4 camera/voice features go live
Exercise Library / assessment data not structured well enough for future AI use	Design data model in Phase 2 with Phase 3 AI consumption in mind, reviewed against Section 2.2 principle
LLM/AI provider cost or reliability issues (Phase 3)	AI Gateway design allows switching or load-balancing across providers without touching application code
Computer vision accuracy/liability concerns (Phase 4)	Position CV feedback as coaching guidance, not medical/injury diagnosis; human trainer remains the authority
Voice AI / outbound calling compliance (Phase 4)	Review telecom consent and do-not-call regulations per operating region before build, not after

15. Approval Ask

Requesting review and sign-off on:

- The four guiding principles in Section 2 (closed-loop model, Member 360, modular monolith, multi-tenancy/events from day one)
- The technology stack in Section 4, including the AI/CV/voice additions for Phase 3-4
- The full Phase 0 → 1 → 2 → 3 → 4 sequencing, module breakdown, and exit criteria in Sections 6-10
- The phase sequencing rationale in Section 13, so expectations are aligned on build order before development starts

Next step upon approval: break Phase 0 down into Epics → Features → User Stories → API specs → DB schema for sprint planning, followed by the same breakdown for Phase 1 once Phase 0 is underway.